

Kernel identity – current-status alignment

TECT verification-first repository · claim A1-KERNEL-IDENTITY

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Purpose and scope

The algebraic theorem and scope are unchanged from v1.0. This issue records the already enacted operator approval and removes the obsolete sign-off action. For $Y > 0$ and $Z < 0$,

$$r + Z|q|^2 + Y|q|^4 = m_{\text{sh}}^2 + Y(|q|^2 - q_\star^2)^2,$$

where $q_\star^2 = -Z/(2Y)$, $m_{\text{sh}}^2 = r - Z^2/(4Y)$, and $r = m_{\text{sh}}^2 + Yq_\star^4$. Thus the zero-momentum and shell masses are distinct.

The current verifier v1.7.0 passes 14/14 overall. Its named `kernel_identity_complete_square` assertion has worst residual 5.68×10^{-14} . The remaining thirteen assertions belong to the shared A1 production-manifest and scalar-positivity audit; they do not enlarge this identity claim.

Devil’s-advocate

1. Confusing $r = K(0)$ with $m_{\text{sh}}^2 = K(q_\star)$ would erase the nonzero shell shift. The displayed identity and the verifier test the two values separately.
2. The identity alone does not imply positivity. Positivity remains a separate conditional statement in A1-SCALAR-ANALYTIC-BRANCH.
3. Updating an approval field could accidentally enlarge the theorem. This issue changes administrative state and naming only; the algebraic scope and T6 tier are unchanged.

Result footer

Result ID:	A1-KERNEL-IDENTITY
Precise statement:	The Brazovskii kernel complete-square identity separates $r=K(0)$ from $m_{\text{sh}}^2=K(q_\star)$.
Scope:	Algebraic identity; $Y>0$, $Z<0$; naming only.
Dependencies:	A1-KERNEL-CONV.
Evidence grade:	ANALYTIC, EXECUTED.
Reproduction command:	<code>python codes/foundations/a1_kernel_checks.py</code>
Expected output:	14/14 PASS overall; named identity assertion PASS.
Falsification gate:	A counterexample to the displayed identity.
Tier before / after:	T6 / T6; administrative alignment only.
No-overclaim statement:	No positivity, production-functional, PDE, measure, vacuum, or physical-selection conclusion.
Next required action:	Preserve the identity and naming firewall.